Carbon Policy in Production Networks

Three Essays on Emissions, Networks, and Climate Policy

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Three essays on firms, emissions, and the production network

1. **Inferring Firms' Emissions from Transactions Data**
   measurement: who emits, when emissions are unobserved? (∼10 min)
2. **Carbon Leakage Within and Across Countries**
   reduced form: does carbon pricing push activity to dirtier firms? (∼10 min)
3. **Carbon Policies in Production Networks**
   structural: how does *who you cover* shape aggregate emissions? (∼25 min)

A common thread: firm-level emissions and the buyer–supplier network, measured in near-universal Belgian administrative data.

Chapter 1

Inferring Firms' Emissions from Transactions Data

with Gert Bijnens · ∼10 minutes

Inferring firms' emissions from transaction data

Question: what can we learn about firms' emissions from **transaction data**?

- **Motivation:** registries & surveys reach only **large installations**, in a handful of countries; **transaction data** covers *every formal firm by design*
- **Idea:** combustion requires **buying fuel** $\rightarrow$ purchases leave traces in the **network**
- **Data:** Belgium — near-universe B2B + **verified EU ETS emissions** as ground truth
- **Result:** a data-driven **elastic-net proxy** reliably tells **emitters from non-emitters** and ranks them by intensity — beating size- and industry-code benchmarks

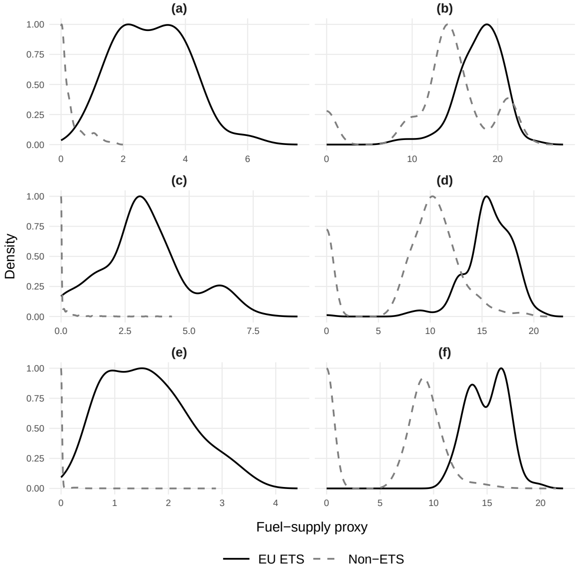
How we build the proxy

- **Train where emissions are observed**:
  - **EU ETS firms** (verified) + non-ETS firms in ∼100%-covered energy sectors (assigned **zero**)
  - the zeros teach the **extensive margin** — and these energy-intensive sectors make it a demanding test
- **Let the data choose suppliers**: regress emissions on **purchases from each supplier**, with an **elastic-net** penalty

$$\text{EN-proxy}_{it} = \sum_{j:|\hat{\beta}_j|>0} \hat{\beta}_j \, x_{ijt} \qquad x_{ijt} = \text{purchases of firm } i \text{ from supplier } j$$

- **Benchmark**: sum purchases from the six **fuel-selling NACE codes** (misses intermediaries; symmetric weights)

What the proxy picks up



- EN selects **612 suppliers** (of ∼22,700), spread across **53 sectors**
- Only **4 of 612** are fuel-seller NACE codes

  ⇒ **data-driven and code-based sets are nearly disjoint**

- The proxy **cleanly separates** emitters from non-emitters

Out-of-sample performance (sector-level cross-fitting)

	Prediction error			Correlation		Extensive margin			
	RMSE	nRMSE	MAPD	Levels	Rank	FPR	TPR	p50	p99
Revenue	190	1.00	**0.71**	0.85	0.53	**1.00**	1.00	0.01	0.39
Elastic Net	218	1.15	0.84	0.85	**0.64**	**0.17**	0.96	0.01	**0.19**
NACE	214	1.13	0.85	0.81	0.52	0.67	0.99	0.01	0.46
Gated Rev.	193	1.01	0.71	0.84	**0.68**	**0.17**	0.96	0.00	0.35

- EN dominates the **extensive margin** (FPR 0.17 vs. revenue 1.00, NACE 0.67)
- Revenue dominates **level accuracy among emitters** (MAPD 0.71)
- **Gated Revenue** — EN classifies, revenue scales — gives the **best of both**

Means over $M = 200$ sector-level CV repeats.

Chapter 1 — takeaways

- **B2B transaction data carry a signal about emissions** beyond balance-sheet variables
- A **cross-fitted elastic-net proxy** beats industry-code classification; a hybrid captures both margins
- **Broader lesson**: combustion $\Rightarrow$ fuel purchases $\Rightarrow$ network traces — a **general logic**, testable wherever transaction data + a partial registry coexist

  $\hookrightarrow$ the same firm-level emissions & network reappear, structurally, in Chapter 3.

Chapter 2

Carbon Leakage Within and Across Countries

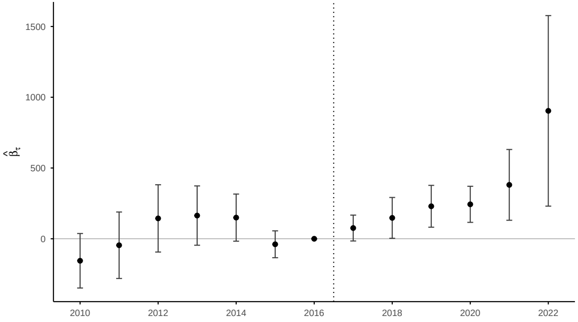
with Gert Bijnens · ∼10 minutes

Carbon leakage within and across countries

Question: does the EU ETS push activity toward *less-regulated* suppliers — **within** Belgium and **across** its borders?

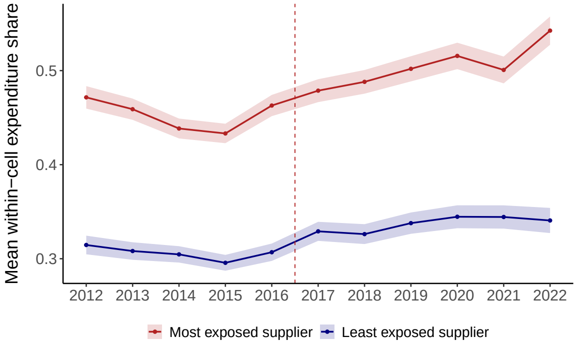
- **Data:** firm-level Belgium — near-universe **B2B network** + firm$\times$product$\times$country **customs** + **EUTL** emissions/allowances
- **Approach:** continuous-treatment DiD — dose = a supplier's predetermined **carbon-cost exposure**; shock = the **2015 MSR**-driven EUA run-up
- **Result:** carbon pricing **does** raise producer prices (more in exposed sectors) — but sourcing **does not move**. **Three margins, three nulls.**
- **Why:** the relative-price gap stays **tiny** even at full pass-through — too small to repay switching

First: the policy does bite on prices



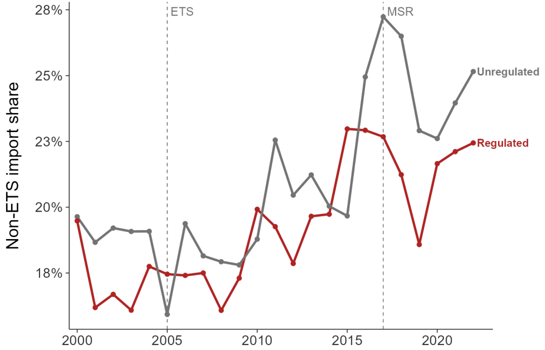
- **Shock**: the 2015 MSR drives EUA **€5 → €80+** — mechanical, predetermined
- **Dose**: supplier's pre-2017 exposure $\omega = \frac{\text{shortage}}{\text{total cost}}$
- Manufacturing prices rise after 2017 and **stay elevated**
- Concentrated in **shortage**-exposed sectors: $+0.012^{**} \rightarrow +0.023^{***}$ per SD (gross emissions: **null**)

Margin 1 — within Belgium: no reallocation



- Expenditure share on **most-** vs. **least-**exposed supplier moves in **parallel** after 2017
- DiD on post$\times$top: $+0.0123^{**}$ (wrong sign) $\rightarrow$ 0.0031 **n.s.** once mechanical attrition is absorbed; not robust under HonestDiD
- **Why**: even at full pass-through the relative-price gap is $< 1\%$ — too small to act on

Margin 2 — across borders: Belgium $\neq$ France



- Replicate Coster–Méjean–di Giovanni on Belgian customs: regulated imports, EU vs. non-EU sources
- Regulated $\times$ non-ETS share is **flat** across 2000–2022

  Intensive margin (value share):
  - Belgium: $+0.0004$ (n.s.)
  - France: $+0.121$ ($p < 0.001$)

  $\Rightarrow$ a common EU carbon price need **not** produce the same leakage everywhere

Chapter 2 — three margins, three nulls

The set of margins with detectable leakage is **empty**.

1. Pass-through is **real** and persistent — more in shortage-exposed sectors
2. **No** within-NACE reallocation: wrong-signed, shrinks to insignificance, not robust
3. **No** international reallocation: flat shares, far below France

Mechanism: the cost shock is small and pass-through partial $\Rightarrow$ the relative-price gap never clears the cost of switching suppliers.

$\hookrightarrow$ Chapters 1–2 measure firms and test reallocation *reduced-form*; Chapter 3 builds the structural model that says *when* reallocation should bind.

Chapter 3

Carbon Policies in Production Networks

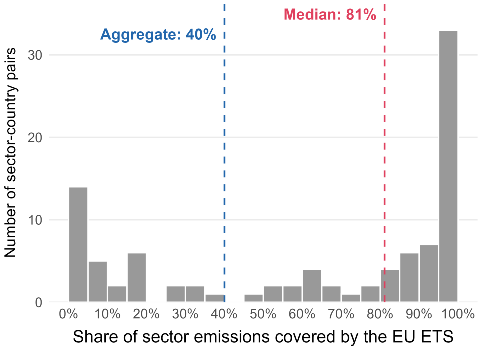
Job Market Paper, with Gert Bijnens · ∼25 minutes

Carbon policies in production networks

Question: a carbon price covers *some* firms, not all — what is its effect on aggregate emissions, and how does it depend on *whom* it covers?

- **Theory:** GE production network, heterogeneous emission intensities + costly **abatement**; carbon price p_z, $0/1$ coverage τ
- **Result:** $\frac{d \log Z}{d p_z}$ = **scale** (≈ 0) + **technique** + **reallocation** — the last a **single covariance** of exposure with network-adjusted intensity Ψ_e
- **Data:** Belgium (B2B + EUTL): under the EU ETS the cut is **mostly abatement**, reallocation a modest amplifier
- **Punchline:** almost all the gain from *whom to cover* is firms' **own** emissions — a constrained regulator just **targets the heaviest emitters**

Carbon pricing is incomplete — and the network matters



EU ETS: priced share within targeted sectors — median ∼81%, aggregate ∼40%.

- ∼75 pricing instruments, ∼**a quarter** of global emissions — coverage **partial & uneven**
- **Three questions**: aggregate effect? dependence on *who* is covered? a single most-valuable firm?
- Emissions are partly **own**, partly **embodied upstream** $\Rightarrow$ exposure *and* the aggregate effect are **network objects**
- Key object: **network-adjusted emission intensity** $\Psi_e := \Psi \overline{\mathcal{E}}$ (scope 1+2+3)

Environment: a production network with abatement

- Closed economy, N producers (CRS, marginal-cost pricing); **nested-CES** inputs (σ_B across sectors, σ_W within); Domar weights λ_i, Leontief $\Psi = (I - \Omega)^{-1}$
- **Policy**: carbon price p_z, coverage $\tau \in \{0, 1\}^N$; emissions a by-product, intensity $\overline{e}_i$
- **Covered firms abate** (effort κ_i, $\alpha_i =$ how hard i is to abate):

$$e_i = (1 - \kappa_i)^{\alpha_i}\, \overline{e}_i \qquad \Longrightarrow \qquad \left.\frac{d \log e_i}{dp_z}\right|_0 = -\,\tau_i\, \alpha_i^2\, \overline{e}_i \ < \ 0$$

- **Key separability**: abatement lowers intensity but leaves the input mix — hence pass-through — unchanged to first order

The carbon price propagates through prices

- Each firm passes its carbon cost into its price; buyers inherit it through inputs:

$$\frac{d\log p_i}{dp_z} = \sum_{j\in\mathcal{N}} \Omega_{ij}\,\frac{d\log p_j}{dp_z} \;+\; \Omega_{iL}\,\frac{d\log w}{dp_z} \;+\; \tau_i\,\overline{e}_i$$

- In matrix form — the **exposure** of every price to the policy:

$$\frac{d\log \mathbf{p}}{dp_z} = \Psi\left(\tau \circ \overline{\mathcal{E}}\right) \;+\; \Psi_{(:,L)}\frac{d\log \lambda_L}{dp_z}$$

$\Rightarrow$ a firm's exposure is its *covered*, network-weighted upstream emission content

Main result: a sufficient-statistic decomposition

Three channels — and **scale** ≈ 0 by Hulten ($\frac{d\log Y}{dp_z}\big|_0 = 0$):

$$\frac{d\log Z}{dp_z}\bigg|_{p_z=0} = \underbrace{-\sum_{i\in\mathcal{N}} \frac{\tau_i z_i}{Z}\,\alpha_i^2\,\bar{e}_i}_{\textbf{technique}\ <\ 0,\ \text{covered firms}} \quad \underbrace{-\,\sigma \sum_{i\in\mathcal{N}\cup\{C\}} \frac{x_i}{Z}\,\mathbb{C}_{\Omega_{(i,:)}}\left(\frac{d\log\mathbf{p}}{dp_z},\ \Psi_e\right)}_{\textbf{reallocation}:\ \text{one covariance}}$$

- Reallocation $=$ one covariance: **exposure** vs. **network-adjusted intensity** Ψ_e
- Its **sign** decides **amplify** ($\mathbb{C} > 0$) vs. **leak** ($\mathbb{C} < 0$); $(1-\sigma)$ sets the **magnitude**
- **Complete coverage**: covariance $\to$ **variance** $\Rightarrow$ reallocation *always* amplifies

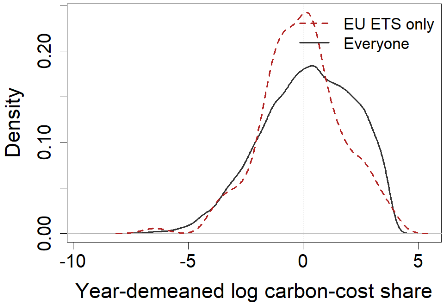
Whom to cover? Emission centrality

- The decomposition is **additively separable** across covered firms $\Rightarrow$ the marginal value of covering firm j alone:

$$\text{Centrality}_j = \underbrace{-\frac{z_j}{Z}\,\alpha_j^2\,\bar{e}_j}_{\textbf{own abatement}} \underbrace{-\,\sigma \sum_{i\in\mathcal{N}\cup\{C\}} \frac{x_i}{Z}\,\mathbb{C}_{\Omega_{(i,:)}}\Big(\Psi_{(:,j)}\,\bar{e}_j,\ \Psi_e\Big)}_{\textbf{network position}}$$

- The object counterfactuals **rank firms on** — and it makes the network's value an *empirical* question

Belgian data: facts & calibration



Distribution of carbon-cost shares at $p_z = 80$.

- Wide **dispersion** in direct intensity ($p_{90}/p_{10} \approx 5.7$ within 4d-year, $\to\sim$30-fold with the imputed tail)
- Direct $\leftrightarrow$ upstream link is **weak**; for **1 in 3** ETS firm-years *upstream* emissions exceed own direct
- Thin within-sector vs. broad across-sector supplier base $\Rightarrow \sigma_W > \sigma_B$

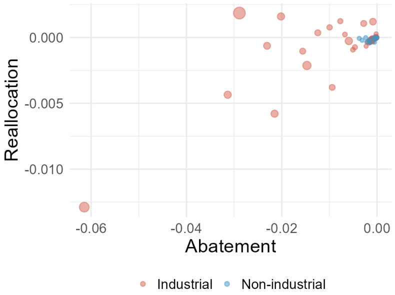
Calibration: 2019 network, EU ETS τ, $p_z = 80\,\text{EUR/t}$; $\sigma_B = 0.1,\ \sigma_W = 2.5,\ \rho = 0.7,\ \alpha = 2$

Result 1 — the EU ETS effect is mostly abatement

Channel	$\Delta \log Z$	share
Total	-0.421	
Scale	-0.000	≈ 0
Technique	-0.371	$\approx 88\%$ ($\sim$7/8)
Composition	-0.049	$\approx 12\%$

- Composition is **negative** $\Rightarrow$ the Belgian network **amplifies** abatement (covered firms sit in cleaner-leaning supply chains)
- But **uneven**: taxing a firm lowers emissions at **$\sim$4 in 5** firms, raises them at the rest (offsetting **$\sim$1/5**)
- For **$\sim$1 in 25** firms, coverage *backfires*: demand shifts to dirtier uncovered rivals

Result 2 — which firms beats how many



Technique vs. composition centrality (corr. 0.74).

- Re-selecting the *same number* of firms toward central ones: $+31.7\%$ vs. ETS
- Expanding to *all* of industry: only $+5.3\%$

$\Rightarrow$ which $\approx 6\times$ how many

But: holding firm count fixed, ranking on **direct emissions z_i alone** recovers **$\sim$94%** of the centrality gain — the network's extra value sits at the *backfiring* margin, not in everyday selection.

- A **transparent decomposition** of targeted carbon pricing: scale (≈ 0), technique, and reallocation — the last a **single covariance** of exposure with network-adjusted intensity Ψ_e
- Empirically (Belgium, EU ETS): the cut is **mostly abatement**; the network **amplifies** modestly
- The network is indispensable for **diagnosis** — measuring the effect and flagging the firms whose coverage **backfires** — but for **selection** a constrained regulator simply **targets the heaviest emitters**

Next: open economy (cross-border leakage — Chapter 2), and the jointly-optimal covered set

Thank you!

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